

Krish Singh

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SUMMARY

CS graduate interested in backend engineering and systems programming, with self-driven projects spanning container runtimes, RAG pipelines, and Linux tooling. Currently supporting IT infrastructure at Invesco. Looking to transition into a backend engineering role.

WORK EXPERIENCE

IT Infrastructure Support

Jan 2025 – Present

Prakruthi Recycling Pvt Ltd | *Deployed at Invesco, Hyderabad*

Part-time → Full-time Jun 2026

- Provision and configure laptops for new hires — setting up software environments, user accounts, and endpoint configurations per company standards.
- Perform hardware triage and replacements, diagnosing faulty equipment and coordinating swap-outs to minimise downtime for Invesco's Hyderabad office.
- Manage IT asset lifecycle including intake, allocation, and decommissioning of hardware across departments.

PROJECTS

ContextQA | *GraphRAG Document Q&A System* | [GitHub](#) | *[In Progress]*

- Building a local-first document Q&A system using GraphRAG — ingests PDFs and docs via Apache Tika, chunks and embeds them into pgvector, and retrieves context-aware answers through a Spring Boot backend.
- Designed the retrieval pipeline to traverse entity relationships (GraphRAG) rather than flat vector similarity, improving answer coherence on multi-document queries.

J-Container | *Rootless Container Runtime* | [GitHub](#) | *Write-up*

- **5x faster startup** (149ms vs 778ms) and **3–6x less memory** (44.7MB vs ~150MB) compared to Docker on equivalent workloads.
- Implemented user namespaces (`CLONE_NEWUSER`), PID isolation, and chroot-based filesystem jails without root privileges, using `newuidmap/newgidmap` for unprivileged UID/GID mapping.

Lumin | *Brightness Controller TUI* | [GitHub](#)

- Built a terminal UI brightness controller in Rust using `ratatui`, targeting Linux/Wayland with a multi-backend architecture: `brightnessctl` for laptop panels, DDC/CI for external monitors, and a GTK4 layer-shell overlay as a universal fallback.
- Implemented a GTK4 overlay backend using `gtk4-layer-shell` with Cairo rendering (`Operator::Clear + Operator::Over`) to simulate per-monitor brightness without hardware access.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript **Exploring:** C, Rust

Backend: Spring Boot, Spring MVC, Hibernate/JPA, REST APIs, Microservices

Databases: PostgreSQL (pgvector), MySQL **DevOps & Tools:** Docker, Git, Bash, Tauri

Frontend: React, Node.js, Monaco Editor

EDUCATION

Osmania University

Master of Science in Computer Science

G. Pulla Reddy Degree College

Bachelor of Science in Mathematics & Computer Science | *GPA: 8.41/10*

Hyderabad, India

Nov 2022 – Dec 2024

Hyderabad, India

Aug 2019 – Sep 2022

CERTIFICATIONS & ADDITIONAL

Certifications: Spring Boot 3, Spring Framework 6 & Hibernate — Chad Darby (Udemy); HTML & CSS: Go Deeper Than Projects (Udemy)

Problem Solving: Active on LeetCode (leetcode.com/u/krish165)

Environment: Linux (Fedora, Arch/Hyprland/Wayland), IntelliJ IDEA

Volunteering: Soham Academy of Human Excellence, Hyderabad — conducted hands-on electronics and robotics practicals for underprivileged government school students to spark early interest in technology.